Lecture 21 2025-11-13

① Nonlinear equations:

(1) Solve $f(x) = 0$ for x (find roots)

(2) Solve $g(x) = x$ for x (find fixed points)

Often one can interchange (1) and (2) through suitable transformation

Methods:

- Bisection method for (1)
- Fixed point method for (2)

$$p_{n+1} = g(p_n) \quad n = 0, 1, 2, \ldots \text{ given } p_0 \text{ starting value}$$

Convergence if (i) $g \in C([a,b])$ with $g(x) \in [a,b]$

(ii) g' exists and $|g'(x)| < 1$ for $x \in (a,b)$

- Newton's method:

$$p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)} \quad \text{for } n \geq 1 \text{ given } p_0 \text{ starting value}$$

Convergence if (i) $f \in C^2([a,b])$

(ii) $f'(p) \neq 0$ at $f(p) = 0$

(iii) p_0 "close enough" to p

- converges quadratically

- Secant method: Newton's method without derivatives

$$p_{n+1} = p_n - \frac{f(p_n)(p_n - p_{n-1})}{f(p_n) - f(p_{n-1})}$$

use $f'(p_n) \approx \frac{f(p_n) - f(p_{n-1})}{p_n - p_{n-1}}$

Methods to accelerate convergence

Order of convergence: $p_n \to p$ $p_n \neq p \ \forall n$. Then it is largest α such that

$$\lim_{n\to\infty} \frac{|p_{n+1}-p|}{|p_n-p|^\alpha} = \lambda > 0$$

* Fixed point method: $\alpha = 1$ if $g'(p) \neq 0$

$\alpha = 2$ if $g'(p) = 0$ and $|g''(x)| \leq M$

* Newton: $\alpha = 2$ if $f'(p) \neq 0$

If $f'(p) = 0$ then p may be a zero of multiplicity $m \geq 1$

i.e. $f(x) = (x-p)^m q(x)$ for $x \neq p$ with $\lim_{x\to p} q(x) \neq 0$

Modify Newton to

1) Define $\mu(x) = \frac{f(x)}{f'(x)}$ and apply Newton to μ

$$p_{n+1} = p_n - \frac{\mu(p_n)}{\mu'(p_n)}$$

2) Use $p_{n+1} = p_n - \frac{m f(p_n)}{f'(p_n)}$ if multiplicity is known.

* <u>Aitken's Δ^2-method</u>: Given $\{p_n\}_{n=1}^{\infty}$ coming from iterative method

$$\hat{p}_n = p_n - \frac{(p_{n+1}-p_n)^2}{p_{n+2}-2p_{n+1}+p_n}$$

$$= p_n - \frac{(\Delta p_n)^2}{\Delta^2 p_n}$$

$$\Delta p_n = p_{n+1} - p_n$$

$$\Delta^2 p_n = \Delta p_{n+1} - \Delta p_n$$

converges faster than the original method

(in the sense $\frac{|\hat{p}_n - p|}{|p_n - p|} \to 0$)

* <u>Steffensen's Method</u>: Accelerate convergence of a linearly convergent fixed point method to quadratic

$p_1^{(0)} = g(p_0)$, $p_2^{(0)} = g(p_1)$, $p_0^{(1)} = \text{Aitken}(p_0^{(0)}, p_1^{(0)}, p_2^{(0)})$

$p_1^{(1)} = g(p_0^{(1)})$

Interpolation:

Given $x_0, \ldots, x_n$ and values at these points $f(x_0) \ldots f(x_n)$
$y_0 \cdots y_n$

Find a polynomial $p(x)$ to approximate $f(x)$

- **Lagrange polynomials:**

$$P(x) = \sum_{k=0}^{n} f(x_k) L_k(x)$$

$$L_k(x) = \prod_{\substack{i=0 \\ i \neq k}}^{n} \frac{x - x_i}{x_k - x_i} \qquad L_k(x_i) = \begin{cases} 1 & \text{if } i = k \\ 0 & \text{if } i \neq k \end{cases}$$

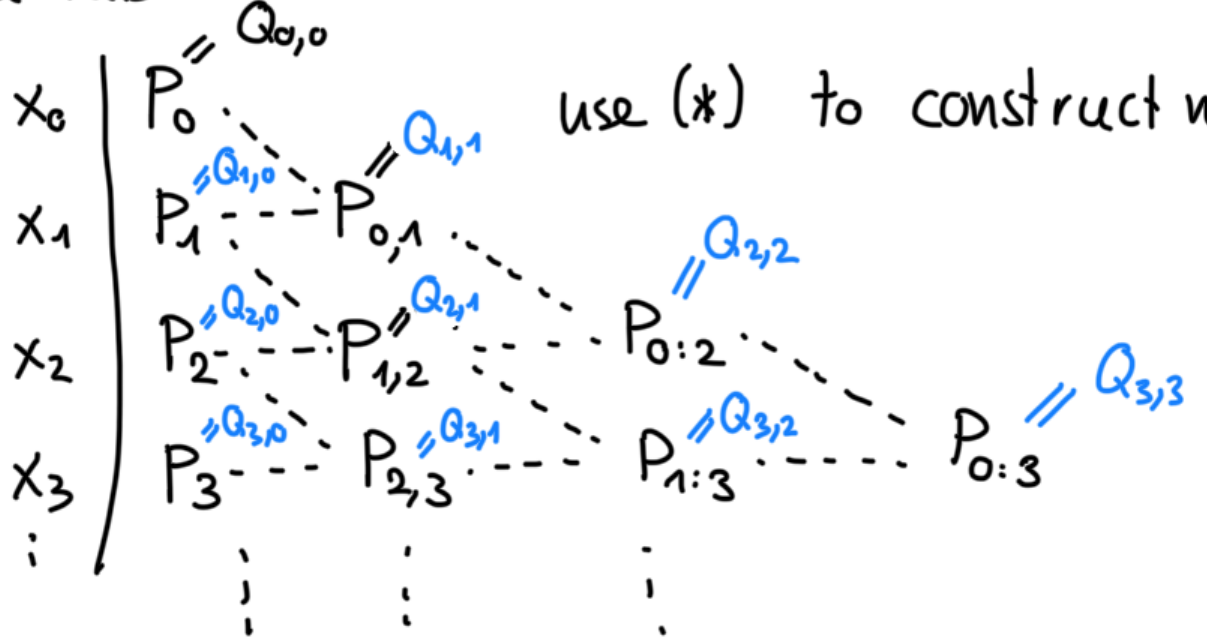
- **Neville's Method:** Recursive construction

$$P(x) = \frac{(x - x_j)\hat{P}_j(x) - (x - x_i)\hat{P}_i(x)}{x_i - x_j} \qquad (*)$$

$\hat{P}_j$ interpolates at all points except for x_j

$$\hat{P}_j(x_k) = \begin{cases} f(x_k) & \text{if } k \neq j \\ ? & \text{if } k = j \end{cases}$$

can build a table



use (*) to construct next column

$P_{i:j}$ interpolates $x_i, x_{i+1}, \ldots, x_j$

$Q_{i,0} = P_i$

Newton Polynomial / divided differences

Construct polynomial iteratively

$$P_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \ldots + a_n(x - x_0) \cdots (x - x_{n-1})$$

use divided differences to determine a_k

Hermite Interpolation: Now we are also given $f'(x_i)$ at x_i, for $i=0,\dots,n$

do divided differences & include derivatives

define $f[x_i, x_i] = f'(x_i)$

"double points"

x_0	$f[x_0]$	
x_0	$f[x_0]$	$f[x_0,x_0] = f'(x_0)$
x_1	⋮	⋮
x_1		⋮
x_2	!	!
x_2	⋮	⋮
i		

Numerical Differentiation & Integration

Numerical Differentiation:

- Approach 1: Approximate f by Lagrange polynomial $p(x)$

  use $p'(x_i)$ as approximation of $f'(x_i)$

- Approach 2: Want $f'(x_i) = \sum_{j=0}^{n} c_j f(x_j) + O(h^p)$ $\quad h = \Delta x_i$

  for some linear combination c_j

  Use Taylor approximation of $f(x_j)$ to determine c_j

  ⇝ fdstencil.m Matlab code

- Approach 3: Richardson Extrapolation

  Use lower order accurate formulas for different mesh sizes h and combine in a clever way

  $N_1(h) = M + K_1 h + K_2 h^2 + K_3 h^3 + \dots$

  approx. formula ↗ ↖ what we want to compute

  Define $N_2(h) = 2N_1(\frac{h}{2}) - N_1(h)$

  $N_2(h) = M + O(h^2)$

- **Numerical Integration / Quadrature**

  - Approximate f by Lagrange polynomial p

and $\int_a^b f(x)\,dx = \int_a^b p(x)\,dx + \text{error}$

trapezoidal, Simpson's, Midpoint rule etc.

- Composite quadrature rules: Do same on subintervals.

Degree of precision: largest m s.t. $\underbrace{\int_a^b p(x)\,dx}_{(*)} = \sum_{j=0}^{n} a_j\, p(x_j)$

for all polynomials $p(x)$ of deg $\le m$

trapezoidal rule: $m = 1$

Simpson's rule: $m = 3$

midpoint rule: $m = 1$

$$(*)\quad \underbrace{\int_a^b (a_0 + a_1 x)\,dx}_{a_0(b-a) + a_1 \frac{b^2-a^2}{2}} \overset{?}{=} \underbrace{\frac{(b-a)}{2}\left(p(a)+p(b)\right)}_{= \frac{(b-a)}{2}\left(a_0 + a_1 a + a_0 + a_1 b\right)}$$

$\vdots$